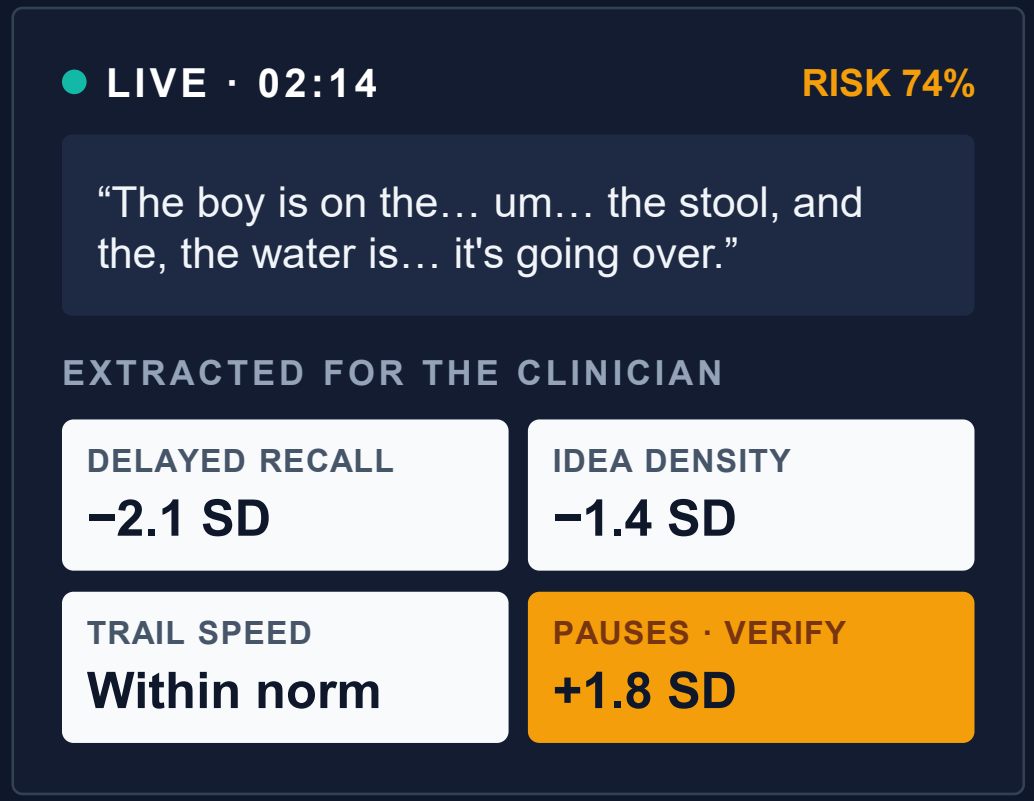
COGNISENSE

MULTIMODAL AI SCREENING FOR EARLY ALZHEIMER'S DISEASE & MCI

Sub-10-minute speech, language and cognitive-task biomarkers, fused into a calibrated, explainable risk tier for the clinician.



Speech biomarkers | **Gamified cognitive tasks** | **Explainable and fair**

Screening aid, not a diagnosis. Research prototype for early triage.

THE DIAGNOSTIC BOTTLENECK

THE HIDDEN CRISIS IN DEMENTIA CARE

Early cognitive decline goes undetected until it is too late, and today's tools cannot scale to every senior.

Silent 20-year window

Amyloid and tau progress for 15–20 years before overt decline, closing the window for new therapies.

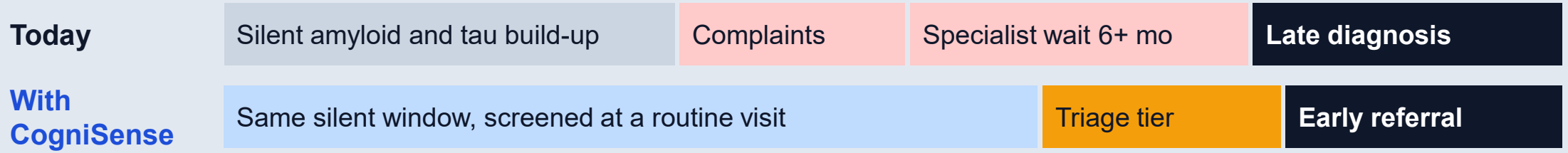
High barrier to entry

PET and CSF cost $3,000–$5,000+; paper batteries take 45–90 min with 6+ month specialist waits.

Inequity and late triage

Over 60% of cases go undiagnosed in primary care, and paper tests carry education bias.

Where the patient journey stalls (illustrative proportions, not measured data)



THE SOLUTION

RAPID MULTIMODAL CLINICAL TRIAGE

CogniSense turns everyday speech and digital interaction into objective biomarkers in under ten minutes.

Dimension	Current primary-care workflow	CogniSense screening layer
Duration	45–90 min paper test; 6-mo wait	**Under 10 minutes, patient-guided on a tablet**
Modality	Subjective paper questionnaires	**Speech acoustics + language + motor/recall**
Access	Needs a neuropsychologist	**Any browser or tablet in the waiting room**
Output	Raw, unadjusted score	**Calibrated risk tier, CI and SHAP explanation**
Integration	Manual paper charting	**One-click HL7 FHIR R4 export to the EHR**
Privacy	Sensitive data stored or outsourced	**Local processing; raw audio wiped**

A screening aid, not a diagnostic verdict: built to triage, not to replace the clinician.

THE PATIENT EXPERIENCE

THE 8-MINUTE PATIENT PROTOCOL

A low-friction, multilingual session that older adults can complete with little help.

1 Consent and audio QC

1.0 MIN

Age, sex and education for norms; microphone SNR check.

2 Picture description

2.5 MIN

Cookie Theft scene captures pauses, syntax and idea density.

3 Memory and fluency

2.5 MIN

Immediate and delayed word recall plus animal fluency.

4 Trail-making

2.0 MIN

Digital trails and taps measure speed, errors and hesitation.

WHAT THE PATIENT SAYS (ILLUSTRATIVE)

“The boy is on the… um… the stool, and the, the water is… it's going over.”

ENGINE CAPTURES (ILLUSTRATIVE)

```
{ "pause_ratio": 0.31,
  "disfluencies": 7,
  "idea_density": 0.42,
  "delayed_recall": "2/10" }
```

SYSTEM ARCHITECTURE

EDGE PIPELINE, EIGHT MODULES

Zero cloud API dependencies: raw biometric audio never leaves the device.

INGEST AND PROCESS →

1 · REACT 18

Patient UI

Tablet-friendly screener on a Zustand state machine.

2 · FASTAPI

Audio QC

Checks SNR, speech length and voice activity first.

3 · FASTER-WHISPER

Local ASR

CPU/edge transcription; verbatim disfluencies kept.

4 · PRESIDIO

Scrub and wipe

Transcript de-identified; raw audio purged after features.

Feature vectors move down to modeling

5 · OPENSMILE

Features

eGeMAPS prosody, spaCy syntax, education-normed z-scores.

6 · EBM · PLATT

Calibrated model

Glass-box risk, SHAP drivers and a conformal band.

7 · AES-GCM 256

Encrypted store

Pseudonymized subject keys and an append-only audit log.

8 · REACT · FHIR R4

EHR export

Clinician reviews, then exports FHIR bundles in one click.

← MODEL AND DELIVER

MACHINE LEARNING PIPELINE

CALIBRATED, LEAKAGE-FREE ML

Every transform is fit inside training folds, and every probability is calibrated before a clinician sees it.

LEAKAGE-FREE BY DESIGN

Fit in folds: scalers and imputers see training data only.

No target leakage: labels and IDs dropped before transforms.

Grouped CV: stratified 5-fold, split by subject.

PIPELINE OUTPUT (ILLUSTRATIVE)

```
{ "features": 47, "cv": "grouped_5fold",
  "calibration": "platt_sigmoid",
  "brier": 0.0866, "tier": "high",
  "ci": [0.68, 0.80], "abstain": false }
```

Engineered feature	Definition	Why it matters
Memory Decay Index	Immediate ÷ delayed recall	Amnestic MCI signal
Cognitive Disfluency Ratio	Silent pause ÷ phonation time	Word-finding difficulty
Executive Slowing Score	Trail errors ÷ completion latency	Executive function
Platt calibration	Sigmoid CalibratedClassifierCV	Honest probabilities

Fit in folds · Subject-grouped · Calibrated

Raw model scores can look 99% sure while guessing. Platt scaling yields a Brier score of 0.0866, so a 65% risk reads as 65%.

VALIDATED SCREENING PERFORMANCE

Prototype results on 5-fold CV and a held-out 20% test set. Screening aid, not a diagnosis.

0.945
Held-out ROC-AUC
5-fold CV: 0.941 ± 0.014.

0.0866
Brier score
Probabilities track risk.

5–25%
Prevalence slider
Dashboard shows real-world PPV.

Metric	5-fold CV	Held-out test	Clinical significance
ROC-AUC	0.941 (±0.014)	**0.945**	Separates impaired from normal cognition
Sensitivity	83.8%	**75.0%**	Triage threshold tuned to 90%+ to limit misses
Specificity	85.9%	**88.5%**	Limits false alarms and needless PET referrals
Precision (PPV)	73.8%	**84.9%**	Balanced research set; see prevalence slider
F1-score	0.782	**0.796**	Balance of precision and recall
Brier score	–	**0.0866**	Probabilities track observed risk

EXPLAINABLE AI

GLASS-BOX EXPLAINABILITY

Biomarker-level attributions instead of an opaque score. Screening aid, not a diagnosis.

CLINICIAN DASHBOARD · CASE #2041 · DEMO DATA ● SCREENED

① CALIBRATED RISK

ELEVATED · HIGH RISK

74.2% CI 68%–80%

Delayed verbal recall deficit with prolonged pauses; motor speed normal. Pattern is consistent with early amnestic MCI.

② TOP RISK DRIVERS (SHAP / EBM)

MEMORY	Delayed recall ratio	−2.1 SD
SPEECH	Pause duration	+1.8 SD
LANGUAGE	Semantic idea density	−1.4 SD
MOTOR	Trail-making speed	Within norm

③ NEXT STEPS

1. Order a formal MoCA
2. Check reversible factors: thyroid, B12, sleep
3. Memory clinic referral for p-tau217 blood test

④ EXPORT FHIR

① Calibrated tier with confidence interval

② Drivers shown as SD from education norms

③ Next steps follow the tier; never a diagnosis

④ One click exports a FHIR R4 bundle

ALGORITHMIC FAIRNESS

FAIR BY DESIGN, PROVEN BY AUDIT

Education, age and sex are checked at every stage, and uncertain cases are never forced into a label.

STEP 1 · NORMALIZE

Education norms

Scores adjusted against bands of 12 years or fewer vs. more than 12.

STEP 2 · AUDIT

Subgroup parity

Sex and age strata (<65, 65–74, 75+) checked for sensitivity and specificity.

STEP 3 · ABSTAIN

Indeterminate tier

Conformal band declines to guess.

Audit	Result	What it shows
Sex proxy probe	**AUROC 0.512**	Sex not recoverable from features
Education proxy	**AUROC 0.487**	No covert education bias
Sex parity	**Male vs. female**	Balanced sensitivity and specificity
Age strata	**<65 / 65–74 / 75+**	Avoids false alarms in normal aging
Uncertain region	**“Indeterminate”**	Re-test or clinician judgment

Proxy AUROC near 0.50 equals chance: demographics cannot be predicted from the model's features.

MULTILINGUAL ACCESSIBILITY

HEALTHCARE FOR EVERY COMMUNITY

Full localization across audio capture, memory word lists and clinician reporting.

English

Stimuli: standard Cookie Theft scene

Recall: CERAD-style word lists

Naming: Boston Naming items

Fluency: animal category task

Hindi (हिन्दी)

Stimuli: culturally adapted scenes

Words: phonetically balanced (फल, पेड़, किताब)

Fluency: localized category tasks

UI: instant language toggle

Marathi (मराठी)

Prompts: localized cognitive tasks

Fluency: culturally resonant categories

Speech: verbatim disfluency tracking

Report: clinician summary localized

Language-invariant core: F0 variation → Pause ratio → Jitter and shimmer → Typed biomarker vector

CLINICAL INTEGRATION

FROM WAITING ROOM TO EHR

Who does what, from patient check-in to the care pathway, in four stages.

	1 · CHECK-IN	2 · EDGE ANALYSIS	3 · CLINICIAN REVIEW	4 · EHR AND CARE PATH
PATIENT Person screened	Checks in; age, sex and education captured	Completes the 8-minute session on a tablet		
EDGE ENGINE Local ASR + ML	Audio QC and consent gate open the session	Local features; raw audio wiped	Tier and SHAP drivers; uncertain = Indeterminate	Builds FHIR R4: Observation, RiskAssessment
CLINICIAN Human decision	Clinic runs on its normal schedule	Tier is ready by exam time	Reviews drivers and normative z-scores	**Exports FHIR; selects care pathway**

LOW · Reassure; annual screen

MODERATE · Lifestyle; retest 6 mo

HIGH · Neurology + p-tau217/PET

LIVE DEMO AND PORTAL

PROTOTYPE DEMO WALKTHROUGH

Production-style patient UX and high-density clinician analytics. Screening aid, not a diagnosis.

/screen
Patient experience

Large touch targets, high contrast

Live microphone level visualizer

Language switch on the fly

/clinic
Clinician portal

Risk tier badge and threshold simulator

Domain radar and z-score bars

Speech pause timeline

/evidence
Evidence hub

Fairness audit and proxy probes

Append-only cryptographic audit log

GDPR right-to-erasure test

LIVE DEMO FLOW

1 · Pick language and consent → **2 · Speak and play the tasks** → **3 · Review tier and SHAP drivers** → **4 · Export FHIR and audit log**

ROADMAP

FROM PROTOTYPE TO CLEARED DEVICE

PHASE 1 · MONTHS 0–3

IRB pilot

IRB-approved clinic pilot

MoCA/MMSE concordance

Hindi and Marathi cohorts

PHASE 2 · MONTHS 3–9

Validation

Plasma p-tau217 truth

Amyloid PET comparison

Multi-site community trials

PHASE 3 · MONTHS 9–18

New signals

Gaze and saccadic latency

Touch and typing dynamics

Longitudinal decline slopes

PHASE 4 · SCALE

Clearance

FDA 510(k) / De Novo

EU MDR Class IIa

SMART-on-FHIR apps

Each phase has one gate: evidence must justify the next step.

SCREEN EARLY. ACT EARLIER.

Fast: multimodal triage in under 8 minutes.

Explainable: calibrated, glass-box risk tiers.

Private: edge processing, audio wiped, fairness-audited.

Team CogniSense

Screening aid, not a diagnosis. Prototype for clinical decision support.

JUDGE Q&A QUICK REFERENCE

1. **Overfitting on small cohorts?**
   Subject-grouped repeated CV, heavy regularization and a frozen held-out split.

2. **PPV in low-prevalence clinics?**
   Triage favors 90%+ sensitivity; the prevalence slider models real PPV.

3. **Demographic or education bias?**
   Education-stratified norms; proxy probes confirm AUROC near 0.50.

4. **HIPAA / GDPR certified?**
   Privacy by design (local ASR, audio wipe, AES-GCM); ready for formal audit.